PROTONS, NEUTRONS AND ELECTRONS

Name: ..... **Date:** .....

Use the rules in the box to fill every gap in the table. Then answer the two questions in full sentences.

Atomic number = number of protons. In a neutral atom, electrons = protons. Neutrons = mass number − atomic number.

COMPLETE THE TABLE

Element	Symbol	Atomic number	Mass number	Protons	Neutrons	Electrons
Hydrogen	H	1	1			
Helium	He	2	4			
Carbon	C		12	6		
Nitrogen		7	14			7
Oxygen	O	8			8	
Fluorine	F	9	19			
Sodium	Na		23	11		
Magnesium		12	24			
Aluminium	Al	13			14	
Chlorine	Cl	17	35			
Potassium	K		39	19		
Calcium	Ca	20	40			

THINK ABOUT IT

An atom has 15 protons and 16 neutrons. Which element is it, and what is its mass number?

One carbon atom has 6 neutrons and another has 8. Are they the same element? Explain.

PROTONS, NEUTRONS AND ELECTRONS

ANSWER KEY

Name: ..... **Date:** .....

COMPLETE THE TABLE

Hydrogen | H | 1 | 1 | 1 | 0 | 1

Helium | He | 2 | 4 | 2 | 2 | 2

Carbon | C | 6 | 12 | 6 | 6 | 6

Nitrogen | N | 7 | 14 | 7 | 7 | 7

Oxygen | O | 8 | 16 | 8 | 8 | 8

Fluorine | F | 9 | 19 | 9 | 10 | 9

Sodium | Na | 11 | 23 | 11 | 12 | 11

Magnesium | Mg | 12 | 24 | 12 | 12 | 12

Aluminium | Al | 13 | 27 | 13 | 14 | 13

Chlorine | Cl | 17 | 35 | 17 | 18 | 17

Potassium | K | 19 | 39 | 19 | 20 | 19

Calcium | Ca | 20 | 40 | 20 | 20 | 20

THINK ABOUT IT

1. Phosphorus (atomic number 15); mass number 15 + 16 = 31. 2. Yes. Both have 6 protons, and the number of protons decides the element. They are isotopes of carbon (carbon-12 and carbon-14).

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